

Factor By Grouping

CONNECT FOUR

PARTNER Game

Also includes factor by grouping guided notes!

Factor by Grouping

Clear Targets:

- I can factor a polynomial with four terms by grouping.

Strategy for Factoring by Grouping:

Step 1: Separate the first two terms from the last two terms.

Step 2: Factor out the GCF from the first two terms, and two terms. After factoring out the GCF from each of the two terms.

Step 3: Combine back into a single expression.

Directions: Factor each polynomial by grouping.

1. $30x^3 - 48x^2 + 5x - 8$

2. $35x^3 - 4$

Factor by Grouping CONNECT FOUR

Partner A:

Partner B:

Partner C:

Partner D:

Partner E:

Partner F:

Partner G:

Partner H:

Partner I:

Partner J:

Partner K:

Partner L:

Partner M:

Partner N:

Partner O:

Partner P:

Partner Q:

Partner R:

Partner S:

Partner T:

Partner U:

Partner V:

Partner W:

Partner X:

Partner Y:

Partner Z:

A	Problem to be Factored	B
	1. $15x^3 + 6x^2 + 20x + 8$	
	2. $9x^3 - 6x^2 - 12x + 8$	
	3. $48x^3 - 56x^2 - 30x + 6$	
	4. $42x^3 + 48x^2 - 7x - 8$	
	5. $4x^3 + x^2 + 8x + 2$	
	6. $25x^3 + 35x^2 - 30x - 42$	
	7. $25x^3 + 20x^2 + 10x + 8$	
	8. $3x^3 - x^2 - 21x + 7$	
	9. $3x^3 - x^2 + 9x - 3$	
	10. $3x^3 - 5x^2 + 21x - 35$	

Name: _____

Date: _____

Factor by Grouping

Clear Targets:

- I can factor a polynomial with four terms by grouping.

Strategy for Factoring by Grouping

Step 1: Separate the first two terms from the last two terms.

Step 2: Factor out the GCF from the first two terms, and then factor the GCF out of the last two terms. After factoring out the GCF from each of the two terms, the remainder should be the same.

Step 3: Combine back into a single expression.

Directions: Factor each polynomial by grouping.

1. $30x^3 - 48x^2 + 5x - 8$

2. $35x^3 - 42x^2 + 20x - 24$

3. $12n^3 + 2n^2 + 42n + 7$

4. $8n^3 - n^2 + 16n - 2$

5. $12x^3 - 9x^2 - 4x + 3$

6. $7x^3 - 14x^2 - 3x + 6$

7. $32m^3 - 56m^2 + 4m - 7$

8. $n^3 + 3n^2 - 4n - 12$

9. $32b^3 - 24b^2 + 12b - 9$

10. $18x^3 - 3x^2 + 12x - 2$

11. $2x^3y^2 - 2x^2y^2 + 3xz - 3z$

12. $12k^3p - 21k^2p - 20kw + 35w$

Factor by Grouping

Clear Targets:

- I can factor a polynomial with four terms by grouping.

Strategy for Factoring by Grouping

Step 1: Separate the first two terms from the last two terms.

Step 2: Factor out the GCF from the first two terms, and then factor the GCF out of the last two terms. After factoring out the GCF from each of the two terms, the remainder should be the same.

Step 3: Combine back into a single expression.

Directions: Factor each polynomial by grouping.

1. $30x^3 - 48x^2 + 5x - 8$

$$(6x^2 + 1)(5x - 8)$$

2. $35x^3 - 42x^2 + 20x - 24$

$$(7x^2 + 4)(5x - 6)$$

3. $12n^3 + 2n^2 + 42n + 7$

$$(2n^2 + 7)(6n + 1)$$

4. $8n^3 - n^2 + 16n - 2$

$$(8n - 1)(n^2 + 2)$$

5. $12x^3 - 9x^2 - 4x + 3$ $(3b^2 - 1)(4b - 3)$	6. $7x^3 - 14x^2 - 3x + 6$ $(7x^2 - 3)(x - 2)$
7. $32m^3 - 56m^2 + 4m - 7$ $(8m^2 + 1)(4m - 7)$	8. $n^3 + 3n^2 - 4n - 12$ $(n^2 - 4)(n + 3)$
9. $32b^3 - 24b^2 + 12b - 9$ $(8b^2 + 3)(4b - 3)$	10. $18x^3 - 3x^2 + 12x - 2$ $(6x - 1)(3x^2 + 2)$
11. $2x^3y^2 - 2x^2y^2 + 3xz - 3z$ $(2x^2y^2 + 3z)(x - 1)$	12. $12k^3p - 21k^2p - 20kw + 35w$ $(3k^2p - 5w)(4k - 7)$

Set Up & Game Play

- **Set-Up:**

- Each partner group will get one list of problems (these are two to a page to reduce the amount of paper used).
- Each student will get a student worksheet to show their work on and a connect four board.
- Optional: Laminate (or put in sheet protectors) connect four boards to use from year to year and use dry erase markers to cross off used circles.

- **Game Play:**

- Students will decide who Partner A is and who Partner B is. One at a time, students choose a problem (they can go in any order, the problems are numbered for easy checking on your part!) from the list and mark off that they have done it in their column (either A or B). Students will factor their chosen problem on the student work paper and then find their final answer on their connect four board and cross it off. Play continues until one partner has 4 in a row horizontally, vertically or diagonally on their board.

- **Differentiation for game play:**

- Only copy one connect four board per partner group and have them mark off their answers on the same board until one person gets 3 in a row. (connecting four is harder when two people are working on the same board). For this one, I would have students initial their circles when crossing it off or use different colored markers/colored pencils to color in the circles.
- Have students choose the problem their partner factors each round.
- Make it a race! Students don't have to wait their turn to factor. Whoever factors to find four in a row first wins! (This doesn't work as well if students pick their own partners, I would suggest pairing students based on ability so you don't have one partner factoring 10 questions and one partner factoring 3 questions!)
 - A variation on this would also have 2 partners working on one team vs. two partners on another team to make the factoring a partner activity and having the two partnerships race each other.

A	Problem to be Factored	B
	1. $15x^3 + 6x^2 + 20x + 8$	
	2. $9x^3 - 6x^2 - 12x + 8$	
	3. $6x^3 + 4x^2 + 9x + 6$	
	4. $42x^3 + 48x^2 - 7x - 8$	
	5. $4x^3 + x^2 + 8x + 2$	
	6. $25x^3 + 35x^2 - 30x - 42$	
	7. $25x^3 + 20x^2 + 10x + 8$	
	8. $3x^3 - x^2 - 21x + 7$	
	9. $3x^3 - x^2 + 9x - 3$	
	10. $3x^3 - 5x^2 + 21x - 35$	
	11. $6x^3 - 8x^2 - 15x + 20$	
	12. $35x^3 + 49x^2 - 25x - 35$	
	13. $x^3 + 6x^2 + 5x + 30$	
	14. $40xy - 35x + 32y - 28$	
	15. $7xy - 28x^2 - y + 4x$	
	16. $15xy + 20x - 24y - 32$	
	17. $6xy - 15x - 4y + 10$	
	18. $5xy + 7x + 5y^3 + 7y^2$	
	19. $10xy + 14x + 5y + 7$	
	20. $xy + 7x - 6y - 42$	
	21. $3xy - x - 15y^2 + 5y$	
	22. $9xy + 24x + 6y^2 + 16y$	
	23. $12xy - 42x^3 + 2y - 7x^2$	
	24. $12xy - 9x^2 - 28ny + 21nx$	
	25. $9xy - 3x^2 - 12ny + 4nx$	
	26. $xy + 6x - ny - 6n$	
	27. $14xy - 10xn^2 + 35ny - 25n^3$	
	28. $24ny - 4n^2 + 18xy - 3xn$	
	29. $20ny - 24n^3 + 15xy - 18xn^2$	
	30. $40ny + 35n^3 + 48xy + 42xn^2$	

A	Problem to be Factored	B
	1. $15x^3 + 6x^2 + 20x + 8$	
	2. $9x^3 - 6x^2 - 12x + 8$	
	3. $6x^3 + 4x^2 + 9x + 6$	
	4. $42x^3 + 48x^2 - 7x - 8$	
	5. $4x^3 + x^2 + 8x + 2$	
	6. $25x^3 + 35x^2 - 30x - 42$	
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	9. $3x^3 - x^2 + 9x - 3$	
	10. $3x^3 - 5x^2 + 21x - 35$	
	11. $6x^3 - 8x^2 - 15x + 20$	
	12. $35x^3 + 49x^2 - 25x - 35$	
	13. $x^3 + 6x^2 + 5x + 30$	
	14. $40xy - 35x + 32y - 28$	
	15. $7xy - 28x^2 - y + 4x$	
	16. $15xy + 20x - 24y - 32$	
	17. $6xy - 15x - 4y + 10$	
	18. $5xy + 7x + 5y^3 + 7y^2$	
	19. $10xy + 14x + 5y + 7$	
	20. $xy + 7x - 6y - 42$	
	21. $3xy - x - 15y^2 + 5y$	
	22. $9xy + 24x + 6y^2 + 16y$	
	23. $12xy - 42x^3 + 2y - 7x^2$	
	24. $12xy - 9x^2 - 28ny + 21nx$	
	25. $9xy - 3x^2 - 12ny + 4nx$	
	26. $xy + 6x - ny - 6n$	
	27. $14xy - 10xn^2 + 35ny - 25n^3$	
	28. $24ny - 4n^2 + 18xy - 3xn$	
	29. $20ny - 24n^3 + 15xy - 18xn^2$	
	30. $40ny + 35n^3 + 48xy + 42xn^2$	

Problem to be Factored	ANSWERS
1. $15x^3 + 6x^2 + 20x + 8$	$(3x^2 + 4)(5x + 2)$
2. $9x^3 - 6x^2 - 12x + 8$	$(3x^2 - 4)(3x - 2)$
3. $6x^3 + 4x^2 + 9x + 6$	$(2x^2 + 3)(3x + 2)$
4. $42x^3 + 48x^2 - 7x - 8$	$(6x^2 - 1)(7x + 8)$
5. $4x^3 + x^2 + 8x + 2$	$(x^2 + 2)(4x + 1)$
6. $25x^3 + 35x^2 - 30x - 42$	$(5x^2 - 6)(5x + 7)$
7. $25x^3 + 20x^2 + 10x + 8$	$(5x^2 + 2)(5x + 4)$
8. $3x^3 - x^2 - 21x + 7$	$(x^2 - 7)(3x - 1)$
9. $3x^3 - x^2 + 9x - 3$	$(x^2 + 3)(3x - 1)$
10. $3x^3 - 5x^2 + 21x - 35$	$(x^2 + 7)(3x - 5)$
11. $6x^3 - 8x^2 - 15x + 20$	$(2x^2 - 5)(3x - 4)$
12. $35x^3 + 49x^2 - 25x - 35$	$(7x^2 - 5)(5x + 7)$
13. $x^3 + 6x^2 + 5x + 30$	$(x^2 + 5)(x + 6)$
14. $40xy - 35x + 32y - 28$	$(5x + 4)(8y - 7)$
15. $7xy - 28x^2 - y + 4x$	$(7x - 1)(y - 4x)$
16. $15xy + 20x - 24y - 32$	$(5x - 8)(3y + 4)$
17. $6xy - 15x - 4y + 10$	$(3x - 2)(2y - 5)$
18. $5xy + 7x + 5y^3 + 7y^2$	$(x + y^2)(5y + 7)$
19. $10xy + 14x + 5y + 7$	$(2x + 1)(5y + 7)$
20. $xy + 7x - 6y - 42$	$(x - 6)(y + 7)$
21. $3xy - x - 15y^2 + 5y$	$(x - 5y)(3y - 1)$
22. $9xy + 24x + 6y^2 + 16y$	$(3x + 2y)(3y + 8)$
23. $12xy - 42x^3 + 2y - 7x^2$	$(6x + 1)(2y - 7x^2)$
24. $12xy - 9x^2 - 28ny + 21nx$	$(3x - 7n)(4y - 3x)$
25. $9xy - 3x^2 - 12ny + 4nx$	$(3x - 4n)(3y - x)$
26. $xy + 6x - ny - 6n$	$(x - n)(y + 6)$
27. $14xy - 10xn^2 + 35ny - 25n^3$	$(2x + 5n)(7y - 5n^2)$
28. $24ny - 4n^2 + 18xy - 3xn$	$(3x + 4n)(6y - n)$
29. $20ny - 24n^3 + 15xy - 18xn^2$	$(3x + 4n)(5y - 6n^2)$
30. $40ny + 35n^3 + 48xy + 42xn^2$	$(6x + 5n)(8y + 7n^2)$

Factor by Grouping Connect Four

Partner A:

Partner B:

$$\begin{pmatrix} 3x^2 + 4 \\ 5x + 2 \end{pmatrix}$$

$$\begin{pmatrix} x^2 - 7 \\ 3x - 1 \end{pmatrix}$$

$$\begin{pmatrix} 5x^2 - 6 \\ 5x + 7 \end{pmatrix}$$

$$\begin{pmatrix} x - 5y \\ 3y - 1 \end{pmatrix}$$

$$\begin{pmatrix} 3x + 4n \\ 5y - 6n^2 \end{pmatrix}$$

$$\begin{pmatrix} 3x - 2 \\ 2y - 5 \end{pmatrix}$$

$$\begin{pmatrix} 3x + 2y \\ 3y + 8 \end{pmatrix}$$

$$\begin{pmatrix} 3x - 7n \\ 4y - 3x \end{pmatrix}$$

$$\begin{pmatrix} 2x^2 + 3 \\ 3x + 2 \end{pmatrix}$$

$$\begin{pmatrix} x - n \\ y + 6 \end{pmatrix}$$

$$\begin{pmatrix} 6x^2 - 1 \\ 7x + 8 \end{pmatrix}$$

$$\begin{pmatrix} 2x + 1 \\ 5y + 7 \end{pmatrix}$$

$$\begin{pmatrix} x^2 + 5 \\ x + 6 \end{pmatrix}$$

$$\begin{pmatrix} x^2 + 2 \\ 4x + 1 \end{pmatrix}$$

$$\begin{pmatrix} x + y^2 \\ 5y + 7 \end{pmatrix}$$

$$\begin{pmatrix} 7x - 1 \\ y - 4x \end{pmatrix}$$

$$\begin{pmatrix} 7x^2 - 5 \\ 5x + 7 \end{pmatrix}$$

$$\begin{pmatrix} 2x + 5n \\ 7y - 5n^2 \end{pmatrix}$$

$$\begin{pmatrix} 6x + 5n \\ 8y + 7n^2 \end{pmatrix}$$

$$\begin{pmatrix} 5x^2 + 2 \\ 5x + 4 \end{pmatrix}$$

$$\begin{pmatrix} x - 6 \\ y + 7 \end{pmatrix}$$

$$\begin{pmatrix} x^2 + 3 \\ 3x - 1 \end{pmatrix}$$

$$\begin{pmatrix} 3x - 4n \\ 3y - x \end{pmatrix}$$

$$\begin{pmatrix} 3x^2 - 4 \\ 3x - 2 \end{pmatrix}$$

$$\begin{pmatrix} x^2 + 7 \\ 3x - 5 \end{pmatrix}$$

$$\begin{pmatrix} 3x + 4n \\ 6y - n \end{pmatrix}$$

$$\begin{pmatrix} 5x + 4 \\ 8y - 7 \end{pmatrix}$$

$$\begin{pmatrix} 5x - 8 \\ 3y + 4 \end{pmatrix}$$

$$\begin{pmatrix} 2x^2 - 5 \\ 3x - 4 \end{pmatrix}$$

$$\begin{pmatrix} 6x + 1 \\ 2y - 7x^2 \end{pmatrix}$$

NAME: _____

DATE: _____ PERIOD: _____

FACTOR BY GROUPING CONNECT FOUR

Directions: Choose a problem from the problem sheet and factor the expression by grouping. Be sure to write the number of the problem you choose in the box where you show your work.

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